Surface Phenomena & Catalysis
Week 5: Zeolites & Carbon Nanotubes

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2302337 — Semester 2/2025

Course Roadmap

Week	Chapter	Topic
1	Ch 1–2	Bridge Module + Computational Perspectives
2	Ch 3	Reactor Design & Catalyst Characterization
3	Ch 4	Transport Limitations in Porous Catalysts
4	Ch 5	Selectivity Engineering
5	**Ch 6**	**Zeolites & Carbon Nanotubes**

How can pore geometry be engineered for molecular-level shape selectivity?

Today's Agenda

Part I: Zeolites *~35 min*

Framework structures, Brønsted acidity, Si/Al ratio, shape selectivity, industrial applications

Part II: Carbon Nanotubes & Diffusion *~30 min*

Chiral indices, functionalization, catalyst supports, configurational vs enhanced Knudsen diffusion

Part III: Summary & Self-Study *~10 min*

Key equations + NB6 self-study guide + course wrap-up

Lecture notes: Chapter 6 — Notebook: NB6

Outline

1 Part I: Zeolites

2 Part II: Carbon Nanotubes & Diffusion

3 Summary & Self-Study

Why Zeolites?

Zeolites are crystalline aluminosilicates with molecule-sized pores that impose **shape selectivity** *on catalytic reactions.*

Building hierarchy:

1. TO_4 tetrahedra (T = Si or Al)
2. Secondary building units (SBUs)
3. Composite building units (sodalite cage, d6r)
4. 3D framework with channels and cavities

Löwenstein's rule

Si/Al ≥ 1 always: no Al–O–Al linkages allowed.

General formula

$$M_{x/n}[Al_xSi_{1-x}O_2] \cdot mH_2O$$

M = charge-balancing cation (valence n)

Key numbers

- >255 unique framework types (IZA)
- Pore diameters: 0.3–1.3 nm
- Surface areas: 300–800 m^2/g
- Used in ∼40% of industrial catalysis

Key Framework Types

Code	Ring	Pore (Å)	Dim.	Si/Al	Application
FAU (Y, USY)	12-MR	7.4	3D	2.5–40	FCC cracking
MFI (ZSM-5)	10-MR	5.1–5.6	3D	15–∞	Xylene isomerisation
CHA (SSZ-13)	8-MR	3.8	3D	5–50	MTO, SCR DeNO$_x$
MOR	12-MR	6.5×7.0	1D	5–20	Alkylation

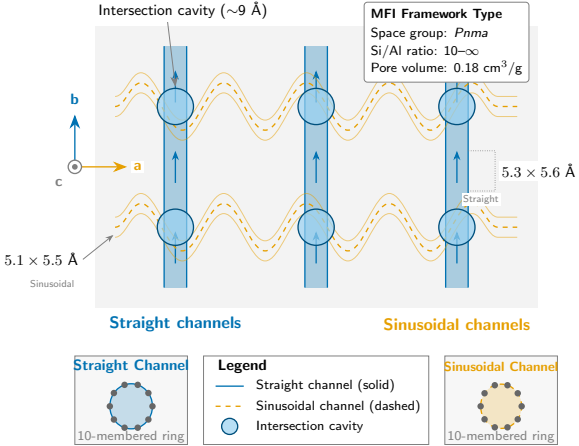
Pore classification

Class	Ring size	Diameter
Small	8-MR	3.0–4.5 Å
Medium	10-MR	4.5–6.0 Å
Large	12-MR	6.0–8.0 Å

Naming convention

12-MR = 12-membered ring
(counts only the T-atoms in the ring)
FAU supercage: ∼13 Å, accessed
through four 12-MR windows (7.4 Å)

MFI (ZSM-5): A Two-Channel System



Straight channels (5.3×5.6 Å, along b) intersect sinusoidal channels (5.1×5.5 Å, along a).
Intersections: ∼9 Å cavities — key for transition-state selectivity.

Brønsted Acidity in Zeolites

When Al^{3+} substitutes for Si^{4+}, one negative charge is created per Al. A proton compensates:

$$\text{Si–O–Si} \xrightarrow{Al^{3+}} \text{Si–}\underline{\text{OH}}\text{–Al}^{-}$$

- The bridging OH is a strong Brønsted acid
- Hammett acidity $H_0 \approx -8$ to -13 (comparable to concentrated H_2SO_4)
- One Brønsted site per framework Al

Acid site types

Type	Origin
Brønsted	Bridging Si–OH–Al
Terminal silanol	Surface Si–OH
Lewis (EFAL)	Tricoord. Al species

EFAL = extra-framework aluminium (from dealumination)

Key distinction

Brønsted sites drive acid catalysis. Lewis sites can **enhance** strength of neighbouring Brønsted sites.

Si/Al Ratio: The Master Design Variable

Acid site density:

$$n_{\text{acid}} = \frac{1}{(1 + \text{Si/Al}) \times M_{\text{avg}}}$$

- Low Si/Al ($\sim$2–5): **many** sites, moderate strength
- High Si/Al ($>$50): **few** sites, strong individual sites
- Higher Si/Al $\Rightarrow$ better hydrothermal stability

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Rule of thumb

FCC (USY): Si/Al ≈ 5–40 (need many sites)
Shape-selective: Si/Al > 25 (need stability)

Si/Al vs properties

	Low Si/Al	**High Si/Al**
Site density	High	Low
Site strength	Moderate	Strong
Hydrophobicity	Low	High
Thermal stab.	Lower	Higher

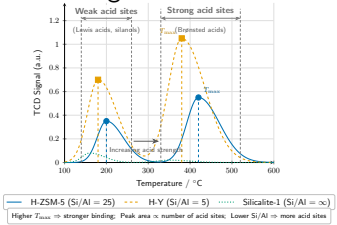
Typical densities

Si/Al	n_{acid} (mmol/g)
1 (max)	∼8–10
5 (H-Y)	∼2.8
25 (ZSM-5)	∼0.64
50	∼0.33

Characterization: NH_3-TPD & Pyridine-FTIR

NH_3-TPD (total acidity):

- Weak sites: 150–300°C
- Strong Brønsted: 300–500°C
- Area $\propto$ total acid sites
- **Cannot** distinguish B vs L



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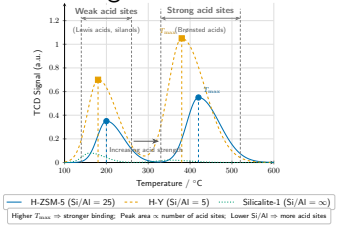

Pyridine-FTIR (B/L speciation):

Band (cm^{-1})	**Assignment**
1545	PyH^+ (Brønsted)
1455	Py–L (Lewis)
1490	Both B + L

Probe molecule limits

- Pyridine (5.3 Å): excluded from CHA
- CO (3.8 Å): probes small pores
- 2,6-DTBPy (7.4 Å): external only

Concept Check — Framework & Acidity

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1. Rank FAU, MFI, and CHA by pore diameter (largest first). Which one admits the smallest molecules?

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2. Two ZSM-5 samples: A has Si/Al = 5, B has Si/Al = 50. Which has more acid sites per gram? Which has stronger individual sites?

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3. Your NH_3-TPD shows a large peak at 200°C but none at 400°C. What can you conclude about the catalyst? Mostly **weak** acid sites (Lewis or silanols); few or no strong Brønsted sites

Three Types of Shape Selectivity

1. Reactant selectivity:
Only molecules smaller than the pore can enter.

Example: n-alkanes enter MFI; branched isomers excluded.

2. Product selectivity:
All products form inside, but only the slimmest can **diffuse out**.

Example: para-xylene escapes ZSM-5; ortho/meta are trapped.

3. Transition-state selectivity:
Only reactions whose TS **fits** the cavity proceed.

Example: bimolecular cracking suppressed in 8-MR cages.

Diagnostic: crystal size test

Selectivity type	S **vs crystal size**
Reactant	Increases
Product	Increases
Transition-state	**Independent**

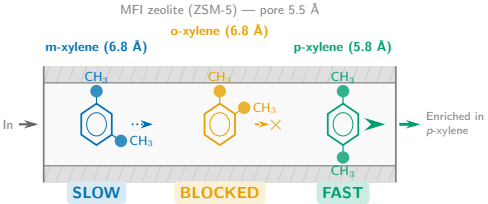
If S does not change with crystal size
$\Rightarrow$ transition-state selectivity.

Selectivity index (Weisz)

$$S_{A/B} = \frac{D_{\text{eff},A}}{D_{\text{eff},B}}$$

In diffusion-limited regime:
observed $S \propto \sqrt{D_A/D_B}$

Product Shape Selectivity: Para-Xylene



Diffusivity ratio: $D_p : D_m : D_o \approx 1000 : 10 : 1$

Diffusivity ratios in ZSM-5

Isomer	D (m^2/s)	D_p/D_i
para (5.8 Å)	10^{-11}	1
meta (6.8 Å)	2×10^{-13}	50
ortho (6.8 Å)	5×10^{-14}	200

Industrial result

>90% para-selectivity vs 24% thermodynamic equilibrium.

>50 Mt/yr production worldwide.
Trapped o/m isomers re-equilibrate inside pores, continuously regenerating para.

Worked Example: Acid Site Density from Si/Al

Problem

H-ZSM-5 with Si/Al = 25. MFI unit cell: $[Al_xSi_{96-x}O_{192}]$.
(a) How many Al per unit cell? (b) Acid site density (mmol/g)? (c) NH_3-TPD gives 0.35 mmol/g. What fraction is accessible?

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(a) $(96 - x)/x = 25 \Rightarrow x = 96/26 = \mathbf{3.7}$ Al per unit cell

(b) $M_{\text{UC}} = 3.7(27) + 92.3(28) + 192(16) = 5756$ g/mol
$n_{\text{acid}} = 3.7/5756 = 6.4 \times 10^{-4}$ mol/g $= \mathbf{0.64}$ mmol/g

(c) Fraction accessible $= 0.35/0.64 = \mathbf{0.55}$ (55%)

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Key insight

Only 55% of framework Al sites are catalytically active. The rest are EFAL Lewis sites, inaccessible interior sites, or framework defects.

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2. FAU has 12-MR pores (7.4 Å) while MFI has 10-MR pores (5.5 Å). Which zeolite would you choose for **reactant** shape selectivity between n-hexane (4.3 Å) and 2,2-dimethylbutane (6.2 Å)?

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Industrial Zeolite Catalysis

FCC — USY (FAU)

- Vacuum gas oil → gasoline
- 500–540°C, 2–4 s contact time
- 12-MR supercage (∼13 Å)
- Carbenium ion cracking mechanism

MTO — SAPO-34 (CHA)

- Methanol → ethylene + propylene
- 8-MR cage traps intermediates; only C_2/C_3 exit (>80% selectivity)
- Hydrocarbon pool mechanism
- >10 Mt/yr capacity (China)

Xylene isomerisation — ZSM-5

- Product shape selectivity
- >90% para (>50 Mt/yr)
- 350–450°C

SCR DeNO$_x$ — Cu-SSZ-13 (CHA)

- $4NO + 4NH_3 + O_2 \rightarrow 4N_2 + 6H_2O$
- Cu^{2+} in d6r cages: sinter-resistant

Post-Synthetic Modification

Dealumination (raises Si/Al)

- Method: steaming at 500–700°C
- Extracts framework Al → EFAL
- Creates intracrystalline mesopores
- Improves hydrothermal stability
- NaY → USY (Si/Al: 2.5 → 5–40)

Desilication (creates mesopores)

- Method: NaOH leaching
- Dissolves Si selectively
- Creates 5–20 nm mesopores ("transport highways")
- Optimal Si/Al: 25–50
  (<15: Al blocks leaching;
  >200: uncontrolled dissolution)

Both methods create **hierarchical zeolites**: micropores for selectivity + mesopores for transport.
Reduces Thiele modulus without changing crystal size.

From Micropores to Nanochannels

What we have so far

- Zeolites exploit **micropore constriction** for shape selectivity
- Si/Al ratio controls acid site density and strength
- Three selectivity types: reactant, product, transition-state
- Industrial impact: FCC, xylene, MTO, SCR

The complementary material

Carbon nanotubes have **atomically smooth** channels where molecules travel **faster** than in free space. Instead of restricting diffusion, CNTs **enhance** it — the opposite design philosophy.

Same Thiele modulus framework, opposite physical origins.

Next: CNT structure, functionalization, and the unified diffusion comparison.

Outline

1 Part I: Zeolites

2 Part II: Carbon Nanotubes & Diffusion

3 Summary & Self-Study

Carbon Nanotube Structure

A CNT is a graphene sheet rolled into a seamless cylinder. The **chiral indices** (n, m) determine diameter and electronic character:

$$d = \frac{a}{\pi}\sqrt{n^2 + nm + m^2}$$

$a = 0.246$ nm (graphene lattice constant)

Three chirality types

- **Armchair** (n, n): always metallic
- **Zigzag** $(n, 0)$: metallic if $n \bmod 3 = 0$
- **Chiral** $(n \neq m)$: mixed

Electronic character

Metallic when $(n - m) \bmod 3 = 0$
$\Rightarrow$ ~1/3 of random (n, m) are metallic.

Metallic CNTs preferred for electrocatalysis (ORR, HER).

Quick calculation

$(10, 0)$ zigzag:
$d = \frac{0.246}{\pi}\sqrt{100} = \mathbf{0.78}$ nm

$(6, 6)$ armchair:
$d = \frac{0.246}{\pi}\sqrt{108} = \mathbf{0.81}$ nm

Single-Walled vs Multi-Walled CNTs

Property	SWCNT	MWCNT
Diameter	0.4–2 nm	5–100 nm
Surface area	300–900 m^2/g	200–400 m^2/g
Synthesis control	More difficult	Easier
Cost	Higher	Lower
Inter-wall spacing	—	∼0.34 nm

Defects as anchoring sites

- Stone–Wales (5-7-7-5 pairs)
- Vacancies (dangling bonds)
- Open tube ends
- Substitutional dopants (N, B)

Defect density: I_D/I_G ratio

- Pristine: $I_D/I_G < 0.1$
  (poor metal dispersion)
- Moderate: $I_D/I_G \approx 0.3$–0.8
  (**optimal** for supports)
- Heavy: $I_D/I_G > 1.5$
  (≈ amorphous carbon)

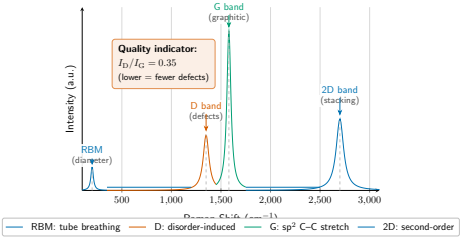
CNT Synthesis: Chemical Vapour Deposition

CVD is the dominant scalable method:

- Carbon source (CH_4, C_2H_2, C_2H_4) decomposes on metal catalyst (Fe, Co, Ni)
- Growth at 600–1000°C
- Catalyst particle size controls tube diameter
- Tip-growth vs base-growth mechanisms

Post-synthesis purification

Acid reflux (HNO_3) removes metal impurities but increases I_D/I_G. Trade-off: purity vs wall integrity.



Raman spectroscopy: G-band (~1580 cm^{-1}, sp^2) vs D-band (~1350 cm^{-1}, defects). I_D/I_G quantifies defect density.

Surface Functionalization & N-Doping

Oxidative functionalization:

- HNO_3 reflux, piranha, $KMnO_4$
- Introduces –COOH, –OH, C=O groups
- Creates anchoring sites for metals
- Improves dispersibility in solvents

Nitrogen doping (4 configurations):

1. **Graphitic N**: substitutional, metallic
2. **Pyridinic N**: at vacancy edges
3. **Pyrrolic N**: in 5-membered ring
4. **Oxidised N**: N–O species

Raman diagnostic

I_D/I_G: 0.35 $\rightarrow$ 0.85 after HNO_3
(2.4$\times$ more defects, but acceptable)

Catalytic significance

Graphitic N $\Rightarrow$ promotes 4e$^-$ ORR pathway
Pyridinic N $\Rightarrow$ Lewis basic sites
N-CNTs approach Pt performance
as metal-free ORR electrocatalysts.

CNTs as Catalyst Supports

Property	CNTs	Activ. C	SiO_2	Al_2O_3
Surface area (m^2/g)	200–900	500–2000	200–800	100–300
Pore structure	Well-defined	Random	Tunable	Mesoporous
Thermal stab. (°C)	$<$600 (air)	$<$400 (air)	$>$1000	$>$1000
Conductivity	High	Moderate	Insulator	Insulator
Metal dispersion	Excellent	Good	Good	Good

Inner vs outer loading

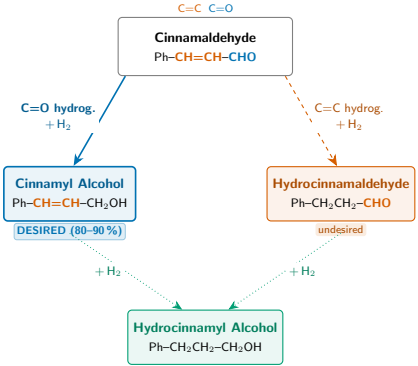
- **Inner**: sinter-resistant (confinement), d-band shift ↓0.1–0.4 eV, limited access for large substrates
- **Outer**: high accessibility, preferred for fast reactions

Why BET area isn't everything

Activated carbon has 2–5× more BET area, yet CNT catalysts often outperform.

Reasons: enhanced transport in CNT channels, better metal dispersion, electronic effects on adsorption energies (d-band centre shift).

Selective Hydrogenation: Cinnamaldehyde



Selectivity Comparison

C=O hydrog. (desired) ▪▪▪ C=C hydrog. ··· Full hydrog.

Pt/CNT: 80–90 % selectivity to cinnamyl alcohol **Pt/C:** 30–50 % selectivity to cinnamyl alcohol

CNT effect: Electronic donation from CNT π-system weakens Pt–C=C interaction, favoring C=O adsorption and hydrogenation.

Why Pt/CNT outperforms Pt/C

- C=O hydrogenation $\rightarrow$ cinnamyl alcohol
- Pt/CNT: **80–90%** selectivity
- Pt/C: only 30–50%

Mechanism (d-band model)

CNT π-system shifts Pt d-band centre **down** $\Rightarrow$ weakens C=C more than C=O $\Rightarrow$ selectivity reversal. CNT modifies $k_{C=O}/k_{C=C}$ by electronic catalyst design.

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3. Why does Pt **inside** a CNT show different selectivity than Pt **outside**?

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3. Why does Pt **inside** a CNT show different selectivity than Pt **outside**? Concave curvature shifts the Pt d-band centre **down** by 0.1–0.4 eV, changing adsorption strengths

Three Diffusion Regimes in Porous Catalysts

Regime	**Pore size**	D_{eff} (m^2/s)	**Controlling factor**
Molecular	>100 nm	$\sim 10^{-5}$	Molecule–molecule collisions
Knudsen	2–100 nm	10^{-6}–10^{-7}	Molecule–wall collisions
Configurational	<2 nm	10^{-10}–10^{-18}	Steric constraints

Classical Knudsen diffusivity

$$D_K = \frac{d_{\text{pore}}}{3}\sqrt{\frac{8RT}{\pi M}}$$

Assumes **diffuse reflection**:
molecules rebound randomly after wall collision.

The nanoconfinement surprise

In zeolites: $D \ll D_K$ (hindered)
In CNTs: $D \gg D_K$ (enhanced)

Same Thiele modulus framework,
opposite deviations from D_K.

Configurational Diffusion in Zeolites

When $d_{\text{pore}} \approx d_{\text{molecule}}$, each molecular "hop" requires overcoming a steric barrier:

$$D_{\text{micro}} = D_0 \exp\left(-\frac{E_D}{RT}\right)$$

$E_D = 10$–50 kJ/mol typical; >100 kJ/mol when molecule $\approx$ pore size

Thiele modulus for zeolite crystals:

$$\phi = R_{\text{crystal}} \sqrt{\frac{k}{D_{\text{micro}}}}$$

Use **crystal radius**, not pellet radius!

(Recall Week 3: same ϕ framework, now with D_{micro})

Why ϕ is often large

D_{micro} is 10^3–$10^8\times$ smaller than $D_K \Rightarrow$ even μm-sized crystals can give $\phi \gg 1$.

Design lever

Reduce crystal size:

- $10\times$ smaller $R \Rightarrow 10\times$ smaller ϕ
- Nano-zeolites (<100 nm) or hierarchical mesopores
- Can raise η: 0.3 $\rightarrow$ 0.9

Worked Example: Zeolite Crystal Thiele Modulus

Problem

H-Y zeolite crystal: $R = 0.5\ \mu$m, $k = 10$ s^{-1} (500°C), $D_{\text{micro}} = 5 \times 10^{-12}$ m^2/s.
(a) Calculate ϕ. (b) Calculate η (sphere). (c) Maximum R for $\eta > 0.9$?

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(a) $\phi = R\sqrt{k/D_{\text{micro}}} = (5 \times 10^{-7})\sqrt{10/(5 \times 10^{-12})} = (5 \times 10^{-7})(1.41 \times 10^6) = \mathbf{0.71}$

(b) $\eta = \dfrac{3}{\phi^2}(\phi \coth \phi - 1) = \dfrac{3}{0.504}(0.71 \times 1.64 - 1) = 5.95 \times 0.163 = \mathbf{0.97}$

(c) For $\eta > 0.9$: need $\phi < 1.32 \Rightarrow R_{\max} = 1.32/(1.41 \times 10^6) = \mathbf{940}$ nm ($0.94\ \mu$m)

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H-Y zeolite crystal: $R = 0.5\ \mu$m, $k = 10$ s^{-1} (500°C), $D_{\text{micro}} = 5 \times 10^{-12}$ m^2/s.
(a) Calculate ϕ. (b) Calculate η (sphere). (c) Maximum R for $\eta > 0.9$?

(a) $\phi = R\sqrt{k/D_{\text{micro}}} = (5 \times 10^{-7})\sqrt{10/(5 \times 10^{-12})} = (5 \times 10^{-7})(1.41 \times 10^6) = \mathbf{0.71}$

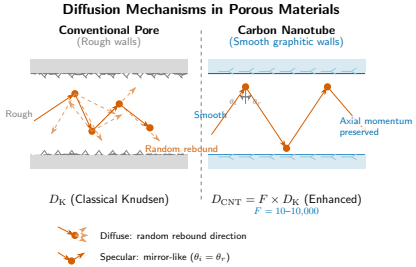
(b) $\eta = \dfrac{3}{\phi^2}(\phi \coth \phi - 1) = \dfrac{3}{0.504}(0.71 \times 1.64 - 1) = 5.95 \times 0.163 = \mathbf{0.97}$

(c) For $\eta > 0.9$: need $\phi < 1.32 \Rightarrow R_{\max} = 1.32/(1.41 \times 10^6) = \mathbf{940}$ nm ($0.94\ \mu$m)

Key insight

At 0.5 μm, the crystal operates near kinetic control ($\eta = 0.97$). But the margin is narrow: $D_{\text{micro}} \sim 10^{-12}$ is $10^7\times$ slower than bulk gas diffusion.

Enhanced Knudsen Diffusion in CNTs



Left: rough pore (diffuse reflection).

Right: CNT (specular reflection).

Atomically smooth sp^2 walls produce **specular reflection**:

$$D_{\text{CNT}} = F \cdot D_K \qquad F = \frac{1+\sigma}{1-\sigma}$$

σ = specularity parameter (0 = diffuse, $\to 1$ = specular)

Typical values

CNT type	σ	F
Pristine SWCNT	0.9–0.98	20–100
Functionalised	0.3–0.5	2–3

$F = 10$–$10{,}000$ reported.

Worked Example: H_2 Diffusion in CNT

Problem

H_2 ($M = 0.002$ kg/mol) in a CNT membrane: $d = 2$ nm, $T = 400$ K, $F = 100$. Reaction: $k = 50$ s^{-1} on confined Pt, $L = 5$ μm. Find (a) D_K, (b) D_{CNT}, (c) ϕ with/without enhancement.

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(a) $D_K = \frac{d}{3}\sqrt{\frac{8RT}{\pi M}} = \frac{2\times10^{-9}}{3} \times 2058 = \mathbf{1.37 \times 10^{-6}}$ m^2/s

(b) $D_{\text{CNT}} = 100 \times 1.37 \times 10^{-6} = \mathbf{1.37 \times 10^{-4}}$ m^2/s

(c) Classical: $\phi = 5 \times 10^{-6}\sqrt{50/(1.37 \times 10^{-6})} = \mathbf{0.030}$ Enhanced: $\phi = \mathbf{0.003}$ $(= \phi_{\text{cl}}/\sqrt{F})$

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Key insight

Both $\phi \ll 1$ ($\eta \approx 1$). For faster reactions or longer tubes, the 100-fold enhancement would be the difference between diffusion-limited and kinetic control.

Concept Check — Diffusion Comparison

Concept Check

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3. Why does functionalising CNT walls **worsen** their transport advantage? Functionalization **roughens** the sp^2 surface, reducing specular reflection ($\sigma \downarrow$, $F \downarrow$)

Two Confinement Extremes: A Unified View

Property	Zeolite micropores	CNT channels
Pore diameter	0.3–1.0 nm	1–50 nm
Diffusion regime	Configurational	Enhanced Knudsen
D vs classical	10^{-3}–$10^{-11}\times$ *lower*	10–$10^{4}\times$ *higher*
Wall character	Rough (ionic framework)	Smooth (sp^2 graphitic)
T-dependence	Arrhenius (E_D)	$D_K \propto T^{0.5}$
Design lever for η	Reduce crystal size	Increase F (wall quality)

The unifying framework

$$\text{Same } \phi = L\sqrt{k/D_{\text{eff}}} \implies \text{Same } \eta(\phi) \text{ curves}$$

Only the magnitude and origin of D_{eff} differ. Zeolites restrict molecules for shape selectivity; CNTs accelerate molecules for enhanced effectiveness.

Outline

1 Part I: Zeolites

2 Part II: Carbon Nanotubes & Diffusion

3 Summary & Self-Study

Key Equations Card

Zeolites:

1. **General formula:**
   $\mathrm{M}_{x/n}[\mathrm{Al}_x\mathrm{Si}_{1-x}\mathrm{O}_2] \cdot m\mathrm{H}_2\mathrm{O}$
2. **Acid site density:**
   $n_{\text{acid}} = 1/[(1 + \mathrm{Si}/\mathrm{Al}) \times M_{\text{avg}}]$
3. **Selectivity index:**
   $S_{A/B} = D_{\text{eff},A}/D_{\text{eff},B}$
4. **Activated diffusion:**
   $D_{\text{micro}} = D_0 \exp(-E_D/RT)$
5. **Zeolite Thiele:**
   $\phi = R_{\text{crystal}}\sqrt{k/D_{\text{micro}}}$

Carbon Nanotubes & Diffusion:

6. **Diameter:** $d = (a/\pi)\sqrt{n^2 + nm + m^2}$
7. **Knudsen:** $D_K = (d/3)\sqrt{8RT/(\pi M)}$
8. **Enhanced Knudsen:** $D_{\text{CNT}} = F \cdot D_K$
9. **Specularity factor:** $F = (1 + \sigma)/(1 - \sigma)$
10. **CNT Thiele:** $\phi_{\text{CNT}} = \phi_{\text{classical}}/\sqrt{F}$

Ten equations — from framework chemistry through the unified Thiele modulus for nanoconfined catalysts.

NB6: Self-Study Exercises

Exercises

Ex	Topic	Priority
1	Zeolite acid site calculation	Essential
2	η vs crystal size plot	Essential
3	Shape selectivity analysis	Essential
4	CNT D_K and F comparison	Essential
5	Zeolite vs CNT Thiele	Essential
6	Raman/TGA data interpretation	Bonus

79 cells | ∼3–4 h |
`https://github.com/ExaPsi/2302337`

Self-study from Chapter 6

Topics not covered in today's lecture:

- MOR and BEA framework details (§6.2)
- Ion exchange and metal loading (§6.4)
- Arc discharge and laser synthesis (§6.7)
- Fischer–Tropsch on Co/CNT (§6.11)

Notebook roadmap

Week 1–2	NB1, NB2, NB3
Week 3	NB4
Week 4	NB5
Week 5	**NB6**

Course Wrap-up

What we have covered (Weeks 1–5)

- **Week 1–2:** Adsorption, Kinetics, Reactors, Characterization (foundations: θ, rate laws, E_a, TOF)
- **Week 3:** Transport limitations (ϕ, η — extended today to nanoconfined materials)
- **Week 4:** Selectivity engineering (parallel/consecutive — applied today to shape selectivity)
- **Week 5:** Zeolites & CNTs (integrates transport + selectivity in real catalyst design)

Exam preparation

- Complete all NB1–NB6 essential exercises
- Review Key Equations Cards (all weeks)
- Practice worked examples end-to-end
- Check Appendix: Equation Summary

The big picture

Surface catalysis is the interplay of:

Adsorption $\times$ **Kinetics** $\times$ **Transport** $\times$ **Selectivity**

Every industrial catalyst design balances these four factors simultaneously.

Thank you and good luck on the exam!